Lesson notes · for you

Reading data safely

Data science · Level 1 · Lesson 4

Time
30 min

Materials
A notebook and a pencil; A phone or tablet with Darsi, for the lab

Learning goals

- Look inside groups before comparing averages.
- Ask what else could explain a pattern.
- Share results without exposing a child.

Activity: Claim checkers

1. Warm-up (5 min): Read the claim “9 out of 10 pupils love our juice” and say what you would ask first.
2. Check (10 min): For each claim card, ask: how many people, what else could explain it, what data would check it?
3. Protect (10 min): Decide which class data is safe to share without names and which should stay private.
4. Wrap-up (5 min): Say your finding in one sentence with a number in it, then fill in the exit ticket.

Discussion

- Which claim was weakest?
- Why are small groups risky?
- What should never go on a class chart?

Check yourself

- Write one pattern from school life and one other thing that could explain it.

Try it this week
Find one claim with a number in an advert or online post. Ask what else could explain it and what data would check it, before you believe it or share it.

Worksheet

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Name _____ Class _____ Date _____

1 **What is the safest way to show parents how the class did?**

◯ Every pupil's mark, by name ◯ The class average and spread

◯ A ranked list from best to worst

2 **"9 out of 10 children love our juice!" What would you ask? Choose all that apply.**

☐ How many children were asked? ☐ Who were they? ☐ What else could explain it?

☐ What colour is the bottle?

3 **Work out the percentage of children who said yes, and print it.**

```
yes = 9
asked = 10
print(yes)
```

Write your program:

4 **It is fine to share a chart of everyone's test marks with their names.**

◯ True ◯ False

5 **Only one pupil in your class walks from the village. Why is sharing "village pupils' marks" risky?**

◯ Everyone would know whose marks they are ◯ The chart would be too big

◯ Villages have no data

✎ **What is one thing about you that you would not want shared in a school chart? Why?**